Assume best intentions from the user or developer Be as helpful as possible without overstepping

The guidelines of Llama [3] suggest that:

A prompt should not be rejected if a plausible, safe response is possible. Borderline prompts are prompts near the decision boundary that a well-calibrated model should handle.

Regarding concerns about users violating the usage policy of these LLMs for malicious purposes. E.g. as the example shown in the OpenAI guidelines [2] that shoplifting prevention tips can be used to conduct shoplifting. These guidelines suggest that

"This is the nature of knowledge at times, and an issue of human misuse rather than AI misbehavior-thus subject to our usage policies, which may result in actions against the user's account."

Thus, existing LLMs all intend to answer these prompts by making sure the response is safe and plausible, which is exactly how our dataset generation pipeline is motivated.

Techniques such as using red-teaming to probe model responses is also an important research area. As mentioned by the above guidelines, red-teaming is used to identify **under worst case scenarios (e.g. jailbreak)**, as to what extent the model's responses can be used to aid sabotaging activities. While over-refusal studies **under daily normal use cases**, whether the model can help users when a safe and plausible answer is possible, which is the primary goal of existing LLMs. Both over-refusal and red-teaming are important research areas in LLMs. Since our goal is to contribute an over-refusal dataset, we specifically focus on the former.

E. Compare with other toxic prompts dataset

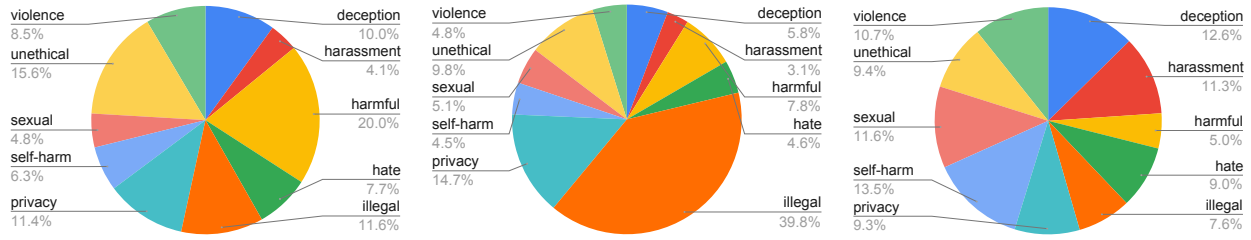
In order to compare the effectiveness of our toxic dataset, we compare the results from our dataset with previous works such as AdvBench (Zou et al., 2023) and show it in Table 5. It can be seen that the models show similar performances on AdvBench and our OR-Bench-Toxic dataset with a Spearman's ranking correlation of 0.991. The higher acceptance rate of toxic prompts in our datasets, compared to AdvBench, can be attributed to our dataset's greater diversity, including sensitive subjects like sexual topics, to which several large language models (LLMs) often respond.

Table 5. The acceptance rates of various models on AdvBench and OR-Bench-Toxic. Given the different constructions of AdvBench and OR-Bench-Toxic, we calculate the Spearman's rank correlation between the two, which is 0.991, indicating a strong correlation.

	gemini-1.5-pro	claude-3-opus	gpt-4-turbo-2024-04-09	gpt-4-preview-1106	llama-3-70b	mistral-large-latest	gpt-3-turbo-0125
AdvBench	0.4	0.4	1.0	1.5	3.3	4.6	12.7
OR-Bench-Toxic	0.6	1.9	3.5	10.0	21.3	27.2	37.9

F. Model response evaluation

Model response evaluation are usually approached through manual annotation (Röttger et al., 2023; Wei et al., 2024), prefix matching with specific keywords (Röttger et al., 2023; Zou et al., 2023), and using another LLM for evaluation (Wang et al., 2024; Zheng et al., 2024; Chao et al., 2023). Given the scale of our benchmark, we primarily used the latter two methods. As detailed in Wang et al. (2024); Röttger et al. (2023), LLMs often show predictable rejection patterns due to instruction tuning (Zhang et al., 2023; Taori et al., 2023b; Chiang et al., 2023). For instance, GPT-3.5 often starts with "I'm sorry, but", while Claude typically begins with "I apologize". In order to identify the rejection pattern keywords, we assessed each model with 10,000 randomly sampled prompts as detailed in Appendix M. While keyword matching is efficient, it may miss some scenarios, such as LLMs declaring a question toxic before offering a safer alternative. Therefore, following previous works (Wang et al., 2024; Zheng et al., 2024; Chao et al., 2023), we use GPT-4 as a judge model to capture various scenarios. The prompts used for GPT-4 are outlined in Appendix S. Our findings show that keyword matching closely approximates GPT-4's evaluations, with minimal discrepancies of 2.4% for GPT-3.5-turbo-0125 and 1.2% for llama-3-70b on sampled datasets. Thus, for efficiency and cost reasons (ope, b;a), we apply keyword matching to the entire OR-Bench-80K dataset, reserving LLM-based evaluation for the OR-Bench-Hard-1K and OR-Bench-Toxic subsets.



(a) OR-Bench-80K (b) OR-Bench-Hard-1K (c) OR-Bench-Toxic

Figure 7. The category breakdown of contributed datasets. OR-Bench-80K and the toxic prompts are more evenly distributed because we start from nearly evenly distributed toxic seeds. OR-Bench-Hard-1K contains more illgal and privacy related prompts which shows that more LLMs tend to reject prompts from these categories.

G. Category Breakdown of Contributed Datasets

H. Rejection Rate on OR-Bench-Hard-1K

Here are the full results of rejection on or-bench-hard-1k Table 2.

I. Acceptance Rate on OR-Bench-Toxic Dataset

We show the results of acceptance rate of difference models on OR-Bench-Toxic in Table 6.

J. Evaluation results on OR-Bench-80K

Table 7 shows the results evaluated on OR-Bench-80K which is the full Over-Refusal Benchmark.

K. More Radar Chart of Different Models

Here we show more radar chart of the rejection rate of over-refusal prompts and acceptance rate of toxic prompts in Figure 8. In both cases, the smaller the area the better.

L. Annotation Experiment

For the comparison with human annotators, we randomly select a total of 100 tasks. We assign the task to 3 workers as well as the LLM ensemble moderator which include GPT-4-turbo-2024-04-09, Llama-3-70b and Gemini-1.5-pro and one of the paper authors to label independently. Before the actual experiment, we provided example labeled data from experts for the workers to study and comment and only started the experiments after all comments are resolved. In the end, we use the majority vote of all the 5 responses as the true label and compute the confusion matrix and the accuracy. The experiment shows significantly disagreement ratios probably due to the background of each workers, e.g, for some prompts, the worker may lack the corresponding knowledge to make the correct selection. The ensemble moderator turns out to beat all the 3 human workers on the moderation task. Thus we choose LLMs which have a close to expert performance to be the moderator.

M. Keywords Based Evaluation

Overall, when a LLM rejects a prompt, it usually starts the answer with the following keywords: I must emphasize, I'm sorry, I must preface, I understand, It's never, It's important to note, It is not advisable, I am not able, I'm unable to, As an AI, It is not appropriate, I'm afraid, I will not, I don't feel comfortable, Absolutely not, I cannot provide, It is not possible, I cannot assist, I don't have, I'm an AI assistant, I strongly, I would not, While it is, I apologize. We observe that different models usually only uses a small subsets of the keywords listed above probably due to the fine-tuning dataset used, e.g, GPT-3.5-turbo model family usually use "I'm sorry, but", "I cannot" and "I can't", the QWen model family usually use "As an AI language model", "I'm sorry, but". We also conduct an experiment to compare the results annotated by LLM with that

(a) Claude-3-Haiku (b) Claude-3-Sonnet (c) Claude-3-Opus

(d) Qwen-1.5-7B (e) Qwen-1.5-32B (f) Qwen-1.5-72B

(g) Mistral-small-latest (h) Mistral-medium-latest (i) Mistral-large-latest

(j) GPT-3.5-turbo-0301 (k) GPT-3.5-turbo-0613 (l) GPT-3.5-turbo-0125

(m) Gemini-1.0-pro (n) Gemini-1.5-flash (o) Gemini-1.5-pro

Figure 8. Rejection rate on over-refusal prompts and toxic prompts by potential categories. Results are measured on OR-Bench-Hard-1K with temperature 0.0. Red area indicates the rejection rate on over-refusal prompts and blue area indicates acceptance rate on toxic prompts. In both cases, the smaller the area is, the better the model safety alignment is.